

**Example 8.3. Inference of a Common Mean of Normal Random Variables.** We observe a collection  $X = (X_1, \dots, X_n)$  of random variables, with an unknown common mean whose value we wish to infer. We assume that given the value of the common mean, the  $X_i$  are normal and independent, with known variances  $\sigma_1^2, \dots, \sigma_n^2$ . In a Bayesian approach to this problem, we model the common mean as a random variable  $\Theta$ , with a given prior. For concreteness, we assume a normal prior, with known mean  $x_0$  and known variance  $\sigma_0^2$ .

Let us note, for future reference, that our model is equivalent to one of the form

$$X_i = \Theta + W_i, \quad i = 1, \dots, n,$$

where the random variables  $\Theta, W_1, \dots, W_n$  are independent and normal, with known means and variances. In particular, for any value  $\theta$ ,

$$\mathbf{E}[W_i] = \mathbf{E}[W_i | \Theta = \theta] = 0, \quad \text{var}(W_i) = \text{var}(X_i | \Theta = \theta) = \sigma_i^2.$$

A model of this type is common in many engineering applications involving several independent measurements of an unknown quantity.

According to our assumptions, we have

$$f_{\Theta}(\theta) = c_1 \cdot \exp \left\{ -\frac{(\theta - x_0)^2}{2\sigma_0^2} \right\},$$

and

$$f_{X|\Theta}(x | \theta) = c_2 \cdot \exp \left\{ -\frac{(x_1 - \theta)^2}{2\sigma_1^2} \right\} \cdots \exp \left\{ -\frac{(x_n - \theta)^2}{2\sigma_n^2} \right\},$$

where  $c_1$  and  $c_2$  are normalizing constants that do not depend on  $\theta$ . We invoke Bayes' rule,

$$f_{\Theta|X}(\theta | x) = \frac{f_{\Theta}(\theta) f_{X|\Theta}(x | \theta)}{\int f_{\Theta}(\theta') f_{X|\Theta}(x | \theta') d\theta'},$$

and note that the numerator term,  $f_{\Theta}(\theta) f_{X|\Theta}(x | \theta)$ , is of the form

$$c_1 c_2 \cdot \exp \left\{ -\sum_{i=1}^n \frac{(x_i - \theta)^2}{2\sigma_i^2} \right\}.$$

After some algebra, which involves completing the square inside the exponent, we find that the numerator is of the form

$$d \cdot \exp \left\{ -\frac{(\theta - m)^2}{2v} \right\},$$

where

$$m = \frac{\sum_{i=1}^n x_i / \sigma_i^2}{\sum_{i=1}^n 1 / \sigma_i^2}, \quad v = \frac{1}{\sum_{i=1}^n 1 / \sigma_i^2},$$

and  $d$  is a constant, which depends on the  $x_i$  but does not depend on  $\theta$ . Since the denominator term in Bayes' rule does not depend on  $\theta$  either, we conclude that the posterior PDF has the form

$$f_{\Theta|X}(\theta|x) = a \cdot \exp \left\{ -\frac{(\theta - m)^2}{2v} \right\},$$

for some normalizing constant  $a$ , which depends on the  $x_i$ , but not on  $\theta$ . We recognize this as a normal PDF, and we conclude that the posterior PDF is normal with mean  $m$  and variance  $v$ .

As a special case suppose that  $\sigma_0^2, \sigma_1^2, \dots, \sigma_n^2$  have a common value  $\sigma^2$ . Then, the posterior PDF of  $\Theta$  is normal with mean and variance

$$m = \frac{x_0 + \dots + x_n}{n+1}, \quad v = \frac{\sigma^2}{n+1},$$

respectively. In this case, the prior mean  $x_0$  acts just as another observation, and contributes equally to determine the posterior mean  $m$  of  $\Theta$ . Notice also that the standard deviation of the posterior PDF of  $\Theta$  tends to zero, at the rough rate of  $1/\sqrt{n}$ , as the number of observations increases.

If the variances  $\sigma_i^2$  are different, the formula for the posterior mean  $m$  is still a weighted average of the  $x_i$ , but with a larger weight on the observations with smaller variance.

The preceding example has the remarkable property that the posterior distribution of  $\Theta$  is in the same family as the prior distribution, namely, the family of normal distributions. This is appealing for two reasons:

- (a) The posterior can be characterized in terms of only two numbers, the mean and the variance.
- (b) The form of the solution opens up the possibility of efficient **recursive inference**. Suppose that after  $X_1, \dots, X_n$  are observed, an additional observation  $X_{n+1}$  is obtained. Instead of solving the inference problem from scratch, we can view  $f_{\Theta|X_1, \dots, X_n}$  as our prior, and use the new observation to obtain the new posterior  $f_{\Theta|X_1, \dots, X_n, X_{n+1}}$ . We may then apply the solution of Example 8.3 to this new problem. It is then plausible (and can be formally established) that the new posterior of  $\Theta$  will have mean

$$\frac{(m/v) + (x_{n+1}/\sigma_{n+1}^2)}{(1/v) + (1/\sigma_{n+1}^2)}$$

and variance

$$\frac{1}{(1/v) + (1/\sigma_{n+1}^2)},$$

where  $m$  and  $v$  are the mean and variance of the old posterior  $f_{\Theta|X_1, \dots, X_n}$ .

This situation where the posterior is in the same family of distributions as the prior is not very common. Besides the normal family, another prominent example involves coin tosses/Bernoulli trials and binomial distributions.